Land screening for agroforestry (ARR) candidacy - Jhabua district, Madhya Pradesh

Independent pre-feasibility screening using only public data | 30 m grid, EPSG:32643 | 2026-10-22

Independent personal project - not affiliated with any organization; pre-feasibility only, not an eligibility d

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REPORT — Land screening for agroforestry (ARR) candidacy, Jhabua district

Independent pre-feasibility screening of candidate land using only public data.

> **Framing — read before any hectare.** This is an **independent personal project** in GIS analysis. It documents a screening methodology; it references the VM0047 methodology (Agriculture, Forestry and Other Land Use — "ARR", Verra VCS vVM0047 v1.1) as a **rubric**, not a certified affiliation, a registration, or a plan to execute a project. **Nothing here is an eligibility determination, a baseline assessment, or a carbon-credit forecast.** The figures below characterise a *screening pool*, not a project boundary.

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1. Executive summary

Question	Result
What is the screening surface?	Jhabua district, Madhya Pradesh, India — **3,433.78 km² / 343,380.6 ha**
At what resolution / CRS?	**30 m**, EPSG:32643, one shared grid (2,933 x 3,130 px), 1 px = 0.09 ha
Stage A (methodology validity)	**324,094.0 ha** (94.4%) methodologically eligible

Question	Result
Stage B (design screens)	**251,292.8 ha** candidate (73.2% of district)
Candidate carrying >=1 **confidence flag**	**197,590.4 ha** (57.6% of district)
Largest exclusion	Protected areas (WDPA), **68,149.1 ha**
QA state	19/19 checks passed, 0 blocking, 0 warnings; raster vs vector area gap **+0.001%**
Test suite	**17/17 passing**
Reproduction	Notebook lab reproduces `reason_codes.tif` and `confidence_flags.tif` **byte-for-byte**

The honest reading of the headline (73.2%): candidate land is a *targeting pool for fieldwork to be conducted later*, not a final project area. That number is high because two deliberate, documented design choices dominate — (1) **the census route keeps cropland** (§4.3(8)(b)/§6.2(2)), which is 60.6% of the district, and (2) the **WDPA screen is conservative** (any polygon excludes, including Reserve/Protected Forest designations, not only IUCN I–IV). Neither is a bug; both are stated in config + docs.

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2. Study area and data

Jhabua district, Madhya Pradesh sits at 74.16–75.01 °E, 22.39–23.24 °N, with a small detached ADM2 component at the southwest (kept in the mask, never silently dissolved). The working grid is EPSG:32643 at 30 m — recalculated against the meters-based geoBoundaries ADM2 boundary layer, not against square-degrees nonsense.

The 30 m contract. Hansen and Copernicus DEM are natively 30 m. Sentinel-2 and WorldCover (10 m) are **aggregated down**, never up: land cover classes use majority voting; Sentinel-2 indices use per-band mean with explicit NaN handling. That means the whole district is **3,815,340 pixels**, each exactly 0.09 ha, on one shared transform/CRS — the single most important constraint enabling every later analysis.

Layer	Source & version	Native	Role in screening
Land cover	**ESA WorldCover 2021 v200**	10 m -> 30 m majority	block built-up/water/wetland; woody-class proxy
Tree cover / loss / gain	**Hansen GFC 2024 v1.12**	30 m -> 30 m nearest	pre-existing woody biomass >=10%; recent removal <=10 yr
Elevation	**Copernicus GLO-30 DEM** (4 tiles)	30 m	Horn slope >20°
Protected areas	**WDPA+WDOECM Oct-2026 (India)**	vector, 188 polys intersecting	blanket protection screen
Boundary	**geoBoundaries 2.0.1 India ADM2**	vector	the district mask & area reference
Imagery (backdrop only)	**Sentinel-2 2024 dry + monsoon composites**	10 m -> 30 m average	rendering, optional scoring

Full provenance (URL, tile, version, SHA256, acquisition date) lives in `data/manifest.json`.

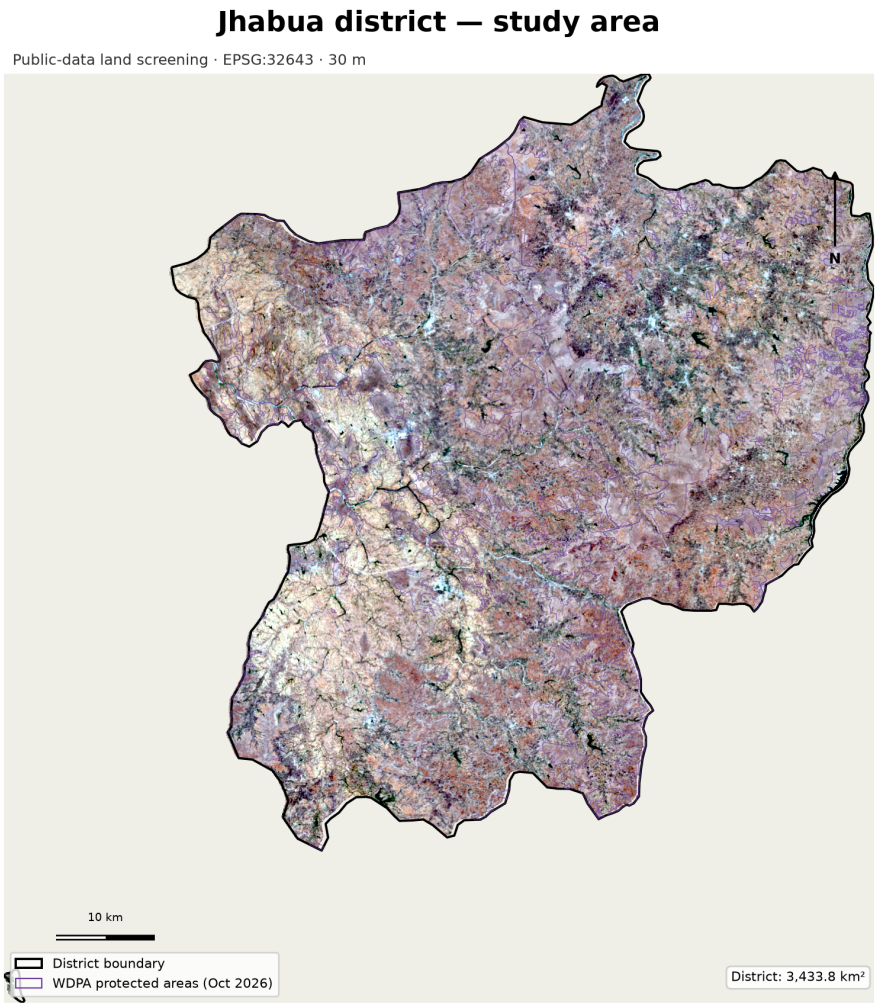


Figure 1 — Study area: Jhabua district polygon overlaid on raw Sentinel-2 2024 dry-season composite; candidate-screening surface in context.

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3. Method

3.1 Pipeline

1. **Ingest.** Six public sources are clipped, mosaic-ed and reprojected onto the shared 30 m grid in EPSG:32643. A missing file **stops the pipeline** with a hard error; it never zero-fills a gap.
2. **QA gate.** 19 **measured** checks: each aligned GeoTIFF is opened and checked for identity (shape, dtype, CRS, nodata), coverage, plausibility (e.g., Hansen loss year <= 2024, so the config is version-aware) and **area reconciliation** — the raster district area must agree with the vector ADM2 area within tolerance. `blockers` fail the run.
3. **Two-stage exclusion waterfall** with **one reason code per pixel** (first failing rule in stage order wins; that pixel's code is final). Stage A is *methodology*, Stage B is *design screening* — deliberately not merged so the two audiences (methodology-eligibility vs project-feasibility) each get their own honest number.

4. **Small patches** re-code to `8` (transaction-cost screen) on what is still eligible after both stages.

3.2 Reason-code contract (canonical, fixed)

code	meaning	treated as
0	**Candidate (code 0)**	the output (never "eligible"); not Stage-A "eligible"
1	Pre-existing woody biomass	Stage A — methodology (§4.3(8)(a)/§6.2(1))
2	Built-up / water / wetland	Stage B — design
3	Cropland excluded	Stage B — **off in census mode (kept 0 ha)**
4	Recent woody removal (<=10 yr)	Stage A — census-route exclusion (§4.4.2(1))
5	Slope > 20°	Stage B — design
6	Protected area	Stage B — WDPA, any polygon
7	Nodata inside district	Stage A — data validity
8	Small patch (< 1.0 ha)	Stage B — transaction cost (removed after Stages A+B)
255	Outside the boundary	excluded from all areas

Attribution rule. For overlapping conditions, each pixel is claimed by the **first failing** screen in stage order, so raw filter areas are supersets, not addends. This makes the pipeline auditable pixel-by-pixel; it also means "protected ha removed" cannot be compared to any independent raw-mask number without re-attributing.

3.3 Data-confidence flags (a separate layer, not exclusions)

`shrubland`, `loss 10-20 yr ago`, and `cropland kept` are **flags** on candidate pixels that mark *uncertainty about the proxy*, not *exclusions* and not *penalties*.

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4. Results

4.1 Waterfall

Step (in order)	Removed (ha)	Remaining (ha)	% of district
District (geoBoundaries ADM2 @30 m)	—	343,380.6	100%
NODATA (data validity, Stage A)	0.0	343,380.6	0%
Pre-existing woody >=10% (Stage A)	19,286.6	324,094.0	94.4%
Recent woody removal <=10 yr (Stage A)	0.0	324,094.0	94.4%
Built-up / water / wetland (Stage B)	4,300.2	319,793.8	93.1%
Cropland exclusion (Stage B) — **off**	0.0	319,793.8	93.1%
Protected areas (WDPA, any polygon)	68,149.1	251,644.7	73.3%
Slope > 20°	144.2	251,500.5	73.2%
Small patches < 1 ha	207.7	**251,292.8**	**73.2% (candidate)**

Two measured zeros, not gaps. The recent-removal screen removes exactly 0 ha because only **12** aligned 30 m pixels inside the district carry any Hansen loss at all (latest loss year **2010**, i.e., far older than the 10-year window). The recent-removal exclusion (§4.4.2(1)) is therefore *not applicable here* on the evidence we have — a fact of the data, not missing data. Hansen covers 2000–2024, so 2025–2026 removals are structurally invisible; that limitation is stated in `docs/limitations.md`.

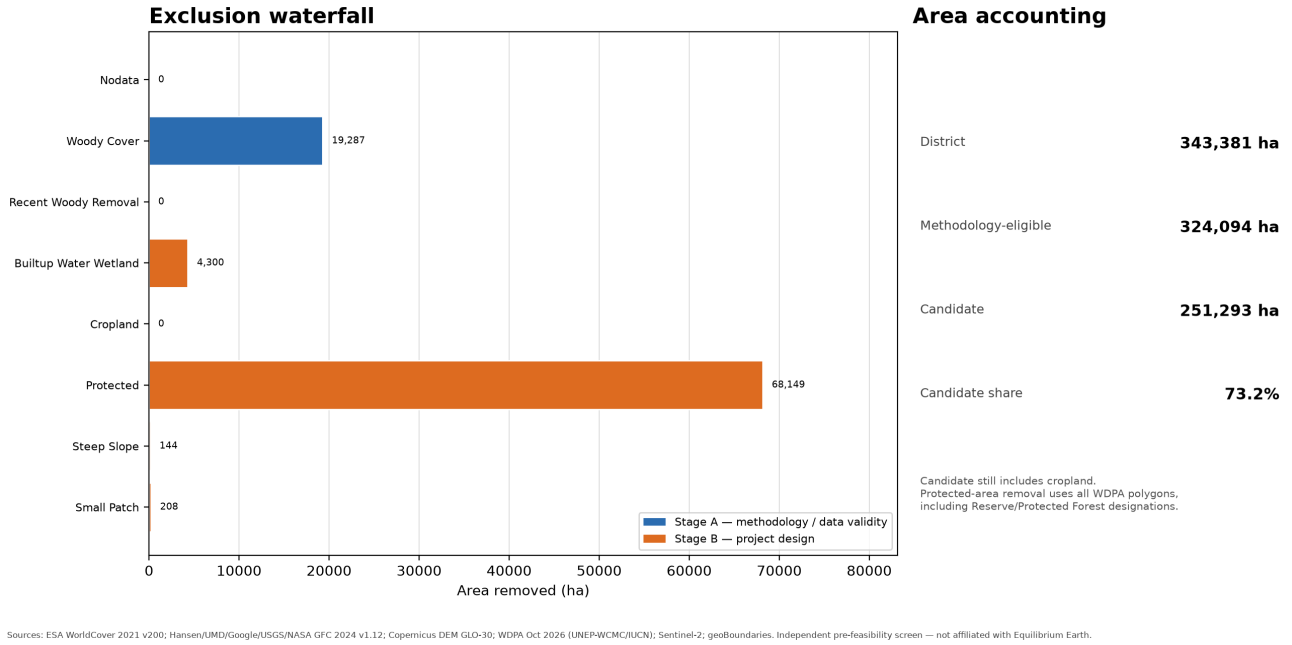


Figure 2 — Stage-A/Stage-B waterfall, same numbers as the table above; rendered directly by the pipeline from `waterfall.csv`.

4.2 Reason codes (the pixel-level audit trail)

code	pixels	ha	% of district
0 Candidate	2,792,142	251,292.8	73.2%
1 Woody	214,296	19,286.6	5.6%
2 Built-up/water/wetland	47,780	4,300.2	1.3%
5 Steep	1,602	144.2	0.04%
6 Protected	757,212	68,149.1	19.8%
8 Small patch	2,308	207.7	0.06%
— Sum (codes 3, 4, 7 absent)	**3,815,340**	**343,380.6**	100%

Every single excluded pixel carries exactly one documented reason; codes 3, 4 and 7 are **absent** because the corresponding screens removed 0 ha — an outcome of the data and config, verified by independent reconciliation.

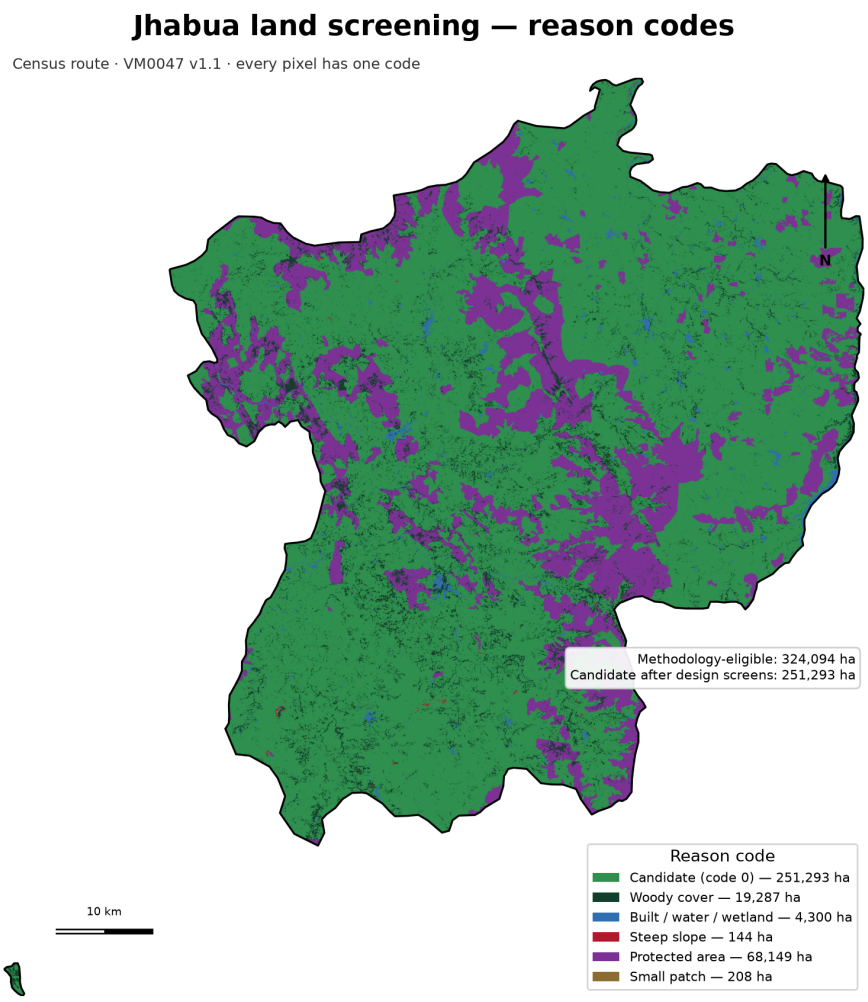


Figure 3 — Per-pixel reason codes. Dark green = Candidate (code 0). Purple = protected; brown-grey = built-up/water; tan = steep; light green = shrubland-flagged candidates.

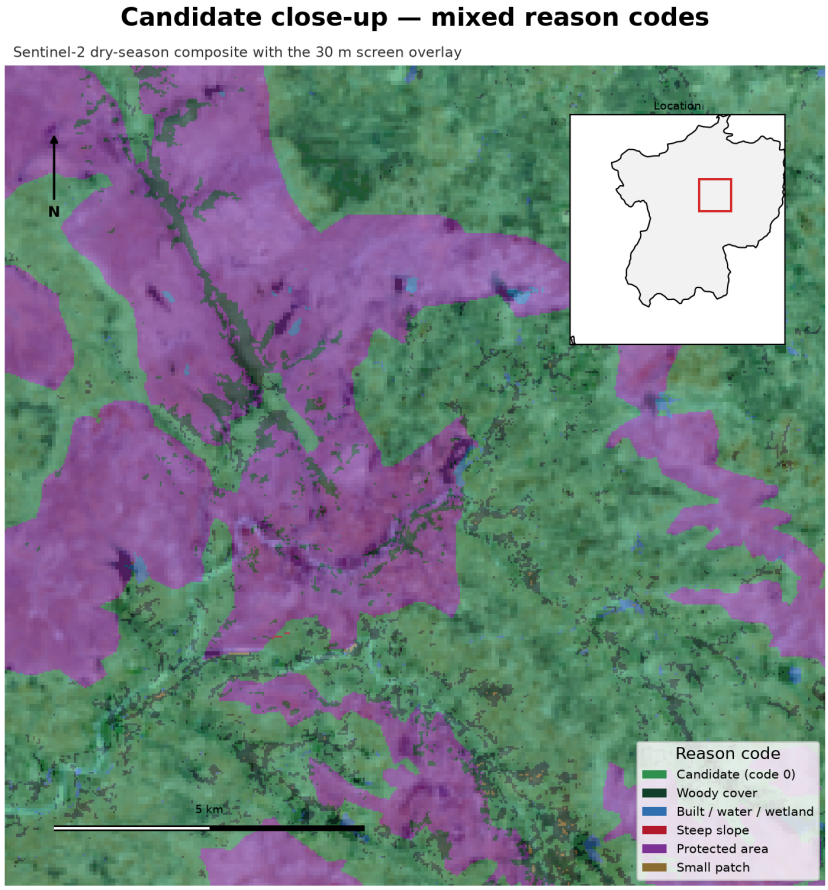


Figure 4 — Zoomed view of candidate land (dark green) with the district boundary; Sentinel-2 2024 dry composite as backdrop.

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5. Analysis

5.1 Why the candidate is 73.2% — land-cover context

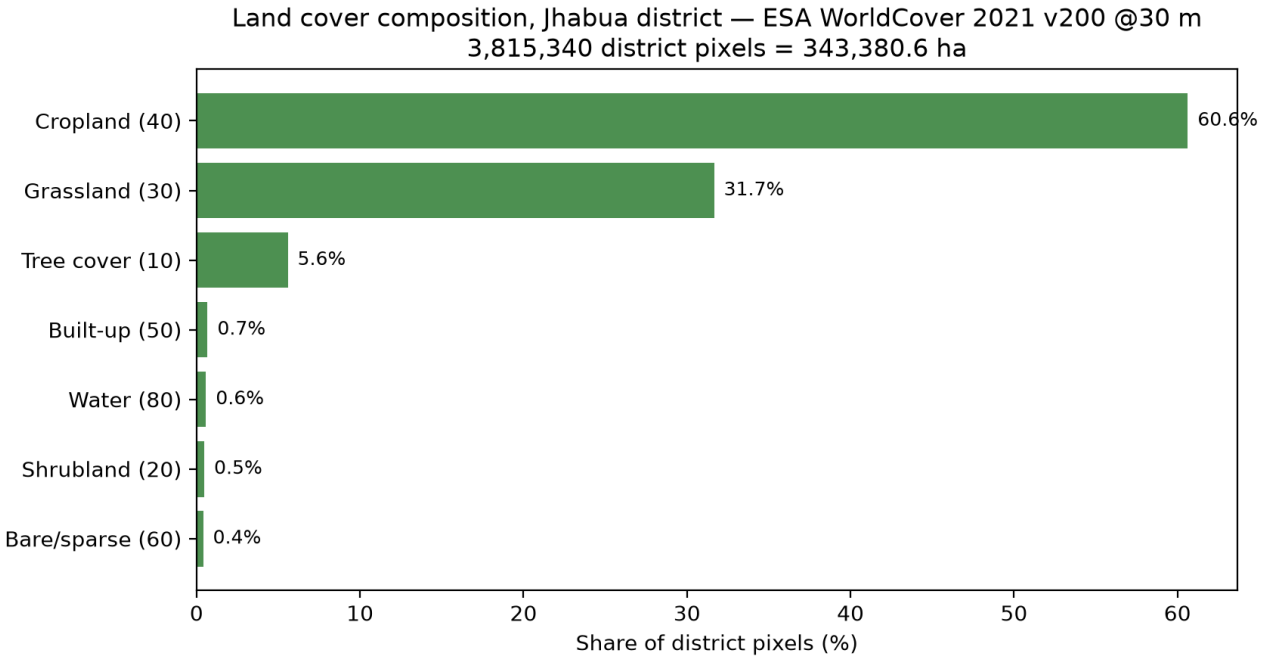


Figure 5 — WorldCover class distribution inside the district (3,815,340 px).

Cropland is **60.6%** of the district and **grassland 31.7%** — together ~92% of the district. Under VM0047's census route (§4.3(8)(b)/§6.2(2)), continuous cropping is **inside** the methodology, not outside it; the design decision `exclude_cropland: false` therefore keeps cropland in the candidate pool, driving the 73.2% headline. Tree cover registers at only 5.6% — and, more importantly, Hansen's >=10% tree canopy occupies just **2,441 px (~219.7 ha)** *inside the district*, which is why the woody screen trims only 19,286.6 ha (5.6%) rather than more.

That is the fastest way to understand this run in one sentence: **Jhabua is a cropland-and-grassland district — a theatre where ARR candidacy lives by keeping ambiguous, low-competition land in the pool and by flagging wherever the proxies cannot vouch for the exclusion they claim.**

5.2 The WDPA screen — conservatism by design

Bounding-box query over India returned **1,376 WDPA+WDOECM polygons**, of which **188** intersect Jhabua. The config excludes **any of them** (including Reserve Forest and Protected Forest designations, not just IUCN I–IV). This is a **conservative design choice**, deliberately trading usable area for defensibility: an IUCN I–IV-only sensitivity would legally re-open a large part of the 68,149.1 ha coded **6**. The over-conservatism is *acknowledged*, not hidden — see `docs/limitations.md` — and a one-line config sensitivity (`wdpa_mode`) is on the follow-up list, not implemented here.

5.3 Raw filters vs waterfall attribution — why the numbers do not add to "68,149 + 144 + ..."

Rule	Raw mask inside district (ha)	Attributed by waterfall (ha)	Difference (ha)
BUILTUP_WATER_WETLAND	4,308.2	4,300.2	+8.0
PROTECTED	73,144.5	68,149.1	+4,995.4
STEEP_SLOPE	905.5	144.2	+761.3
WOODY	19,286.6	19,286.6	0.0

Because Stage-B raw masks **overlap**, the first-fails-wins rule credits each pixel to the earliest screen in `stage_a_order + stage_b_order` that rejects it. Steep pixels are (correctly) claimed 905.49 ha by *slope* <= raw, but the waterfall only attributes 144.18 ha to slope because the rest were already claimed by *protected* or another rule earlier in stage order. **This is not an arithmetic error; it is the attribute of a decision-compatible audit trail**, and the notebook lab reproduces this exactly (byte-for-byte identity with `reason_codes.tif`).

5.4 Fragmentation — what "83% of the workable area is one patch" means

On the *pre-removal* candidate surface (after Stages A+B, before code 8), there were **1,051** patches; after removing patches < **1.0 ha** (connectivity 8), **169** survive, losing 207.7 ha (0.06% of district). Crucially:

- **1 patch holds 244,416.0 ha** — 97.3% of the entire candidate is one contiguous component covering the district's cropland heartland.
- The **median** of the 169 patches is only **3.33 ha** — a long tail of small pockets.

The practical implication: the candidate is **not confetti**. It is one big field (of which cropland is the habitat), so transaction cost in this screening pool will be **community-aggregation-led** rather than patch-detection-led. That's a community engagement fact, not a GIS fact, but the GIS work is what reveals it.

5.5 Confidence flags — what the pipeline flags rather than excludes

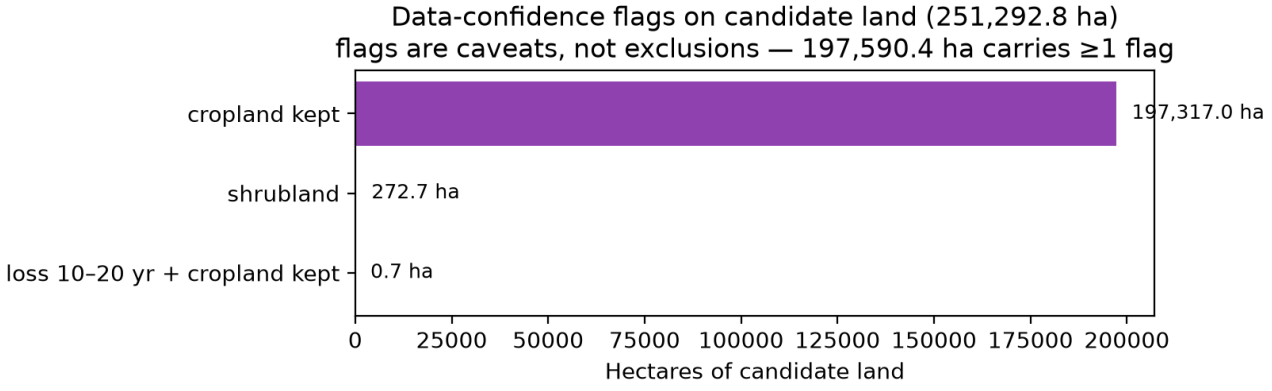


Figure 6 — Flag composition on the candidate surface. Flags mark uncertainty, not exclusion; several can co-occur.

Flag set (candidate land only)	ha	% of candidate
`cropland kept`	197,317.0	78.5%
`shrubland`	272.7	0.11%
`loss 10-20 yr ago` (with cropland kept)	0.7	< 0.01%

Flag set (candidate land only)	ha	% of candidate
>=1 flag	**197,590.4**	**78.6%**

197,317.0 ha of the candidate is flagged **because the census route deliberately keeps cropland** — a *design* caveat (the cropland mask is not validated), not an exclusion. Shrubland (272.7 ha) is flagged because woody *biomass* can register at a canopy-cover proxy even when thicket biomass exceeds the threshold. Only **53.5 ha** of candidate (excluding the flag) is un-flagged free cropland-and-grassland. The rest of the candidate (251,292.8 - 197,590.4 = 53,702.4 ha) carries no flag — mostly grassland with a clear Hansen canopy proxy. That is a **clean pool**, but not a *project-ready* pool — tenure, community, and safeguards live outside the screen.

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6. Data quality and verification

- **QA gate: 19/19 checks pass, 0 blocking, 0 warnings.** Area reconciliation between raster and boundary: **raster 343,380.6 ha vs reference 3,433.78 km² -> gap +0.001%**.
- **Test suite: 17/17 passing** (`python -m pytest -q`).
- **Hands-on reproduction:** `notebooks/learn_by_hand.ipynb` re-implements the entire screen from scratch — grid contract, pixel area, woody rules, stage order, patch pass, flags — and asserts **byte-for-byte equality** with `reason_codes.tif` and `confidence_flags.tif` (8/8 capstone asserts pass).
- **Three production bugs found and fixed during the build**, each now guarded:
  1. **Wrong Hansen tile band** (`20N_070E` covers 20°N–10°N — the first download was wrong; Jhabua is at 22.4–23.2°N and needs `30N_070E`). Fixed by coordinate-band checking before ingest.
  2. ** silent NaN propagation** in Sentinel-2 reads (`src_nodata=np.nan`; NaN is not data).
  3. **Matplotlib `imshow` extent order** is `(left, right, bottom, top)`; the naive `(left, bottom, right, top)` renders a *visually plausible, silently wrong* map — a real, common, and worth-writing-down bug.

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7. Limitations — state these before quoting a hectare

- **30 m grid.** ±0.09 ha per-pixel quantization; for targeting and triage, not for registration, boundaries or measurement.
- **Proxies, not biomass.** Tree canopy (Hansen) + WorldCover classes are proxies for "pre-existing woody biomass". Shrubs and thickets can exceed 10% woody biomass without triggering either.
- **Hansen ends 2024**; the 2025–2026 window is invisible. The recent-removal zero above is a measured zero, not an absence of evidence for 2025–2026.
- **WDPA over-exclude.** Any-polygon designation screen is intentionally conservative; IUCN-only sensitivity pending.
- **No tenure/FPIC/PESA–FRA/safeguard/baseline/additionality/leakage/fieldwork.** This screen concerns itself with *spatial eligibility surrogates only.*
- **The district boundary is ADM2-derived** and includes the detached southwestern fragment; a registry-grade boundary would be needed for any official use.

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8. Reproduce

```
python -m pip install -r requirements.txt
python src/ingest.py config/jhabua.yaml         # builds the aligned 30 m stack
python src/qaqc.py --check config/jhabua.yaml   # 19 measured checks; hard-fails on error
python src/screening.py config/jhabua.yaml      # waterfall + reason_codes + flags
python scripts/make_maps.py config/jhabua.yaml # the 4 portfolio maps
python -m pytest -q                             # 17 tests
```

Configure everything in `config/jhabua.yaml` — thresholds, stage orders, classes, cropland/patch/slope screens — there is **no magic number** in the pipeline code. Raw source provenance (URL, tile, version, SHA256, acquisition date) lives in `data/manifest.json` and is never redistributed.

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9. References

- Verra **vVM0047 v1.1** (ARR, 2025-05-14) — methodological rubric only. PDF in repo with SHA256.
- Hansen M.C. et al. **Global Forest Change 2024 v1.12** (30 m, 2000–2024).
- ESA **WorldCover 2021 v200** at 10 m.
- ESA **Copernicus GLO-30 DEM**.
- **Protected Planet WDPA + WDOECM Oct-2026** (India) — disposition: attribution + no-redistribution.
- **geoBoundaries ADM2** (India, 2.0.1).
- ESA **Sentinel-2** L2A composites (2024 dry + monsoon median).
- Software documentation: `rasterio`, `geopandas`, `numpy`, `scipy.ndimage`, `matplotlib`.

Details: `docs/methodology.md` (method interpretation), `docs/limitations.md` (what can't be claimed), `docs/case_study.md` (portfolio narrative + CV bullets), `learning.md` (the 12-module GIS training manual), `notebooks/learn_by_hand.ipynb` (the byte-exact reproduction lab).